

Run 04 | Fear of failure

India

IV + controls

Target Fear of failure prevents starting a business; binary 1 vs. 0
Sample 27,276 respondents **Coverage** 1 country; 2015–2022
Positive cases 12,009 (44.03%) **Case time** 1m 07s
Train / val. / test 18,549 / 3,273 / 5,454 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business exit, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.7374). Highest F1: Random Forest (0.6629). For Random Forest, adding controls changed ROC-AUC by +0.0064 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.7374	0.6926	62.61	54.96	83.51	0.6629	2s
H2O GBM	0.7235	0.6715	62.63	55.07	82.09	0.6592	7s
CatBoost	0.7226	0.6681	62.89	55.45	79.84	0.6545	23s
XGBoost	0.7212	0.6694	61.17	53.76	84.30	0.6565	1s
Ridge	0.6820	0.6249	54.77	49.25	90.05	0.6367	12s
Logistic Regression	0.6820	0.6245	55.65	49.79	89.13	0.6389	<1s
Elastic Net	0.6812	0.6246	52.00	47.71	94.29	0.6336	6s
Lasso	0.6811	0.6244	53.26	48.38	92.25	0.6348	6s
Decision Tree	0.6588	0.5825	58.10	51.58	78.84	0.6236	<1s
Naive Bayes	0.6409	0.5666	52.51	47.89	89.13	0.6230	7s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Age	357.77	10.24
2	Household size	242.23	6.93
3	Occupational status	218.51	6.25
4	Good career choice	184.85	5.29
5	Household income	159.60	4.57
6	Education attainment	158.15	4.53
7	Media coverage	150.04	4.29
8	Opportunity perception	138.67	3.97

Share is the percentage of total forest impurity importance across all 64 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Investment environment	47.59	1.36
2	Health	39.68	1.14
3	Safety and security	37.16	1.06

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 4 batch, 29 September 2026; 04_obj4_india_iv_cv_fearfail_2015_2022. Scores and importance are taken directly from saved CSV outputs.